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Student Innovative Project 2025 Report

**Point Switch Functionality Detection in Railroads**

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**BONAFIDE CERTIFICATE**

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## 1. Abstract

Railway point switches are critical for route selection in rail networks. Failures in their operation can lead to catastrophic derailments. This project introduces a **sensor-based, wireless-enabled detection system** to monitor the functionality of railway point switches in real-time. The core of this system is an **inductive proximity sensor** that checks the alignment of a loop track with the main track. If alignment is proper, the sensor triggers a **transmitter unit**, which sends the status via **LoRa communication** to a **receiver installed in the locomotive cabin**. Based on the received signal, the system notifies the loco pilot through messages and LED indicators, reducing reliance on manual inspection and enhancing safety.

## 2. Introduction

### 2.1 Background

Train safety largely depends on accurate route switching. Manual errors and mechanical faults in point switches have caused derailments worldwide. Automating the detection of switch alignment helps ensure safe routing.

### 2.2 What is a Point Switch?

A **point switch** is a crucial mechanical assembly in railway systems that enables trains to move from one track to another. It functions by altering the path of the train wheels through the precise movement of rails. The switch consists of two main movable rails, known as **switch rails**, which shift sideways to align with one of two fixed rail paths. By changing the position of these switch rails, the train is either kept on the main line or diverted onto a different track.

This movement is typically powered by an **actuator**, which can be manually operated by a lever or controlled remotely using electric or hydraulic systems. For safety, a **detection mechanism** ensures the rails are correctly positioned before a train passes. In modern systems, this is often done using sensors that send signals to a control centre, confirming whether the switch is properly set and locked in place.

The point switch plays a vital role in managing train traffic, especially at junctions, stations, and yards, ensuring smooth and safe transitions between tracks.

### **3. Problem Statement**

Railway networks are among the most critical components of transportation infrastructure. Within these systems, railway point switches (also known as turnouts) play a vital role in directing trains from one track to another. These switches are especially common at junctions, depots, and crossovers, where the routing of a train needs to change dynamically depending on its destination or traffic conditions.

However, despite advances in railway technology, many switching systems still depend heavily on manual operations and verification processes, particularly in regions or zones that lack modernized or automated infrastructure. The actuation of the switch rails might be controlled remotely via an electrical system, but the confirmation of whether the switch has been properly aligned is often verified manually—either by trackside personnel or through delayed signalling relays. This creates a significant gap in safety and efficiency, especially in high-traffic or high-speed rail environments.

#### **Key Issues with Manual Verification:**

##### **1. Prone to Human Error**

Manual verification introduces the possibility of mistakes due to fatigue, miscommunication, misjudgement, or oversight by railway personnel. In fast-paced operations, even a minor lapse can lead to the switch rails not being properly aligned or locked into position. If a train approaches such a misaligned switch at high speed, the risk of derailment becomes substantial.

##### **2. Delayed Status Updates to the Train Pilot**

In traditional systems, the train pilot receives information about the track alignment via control signals or station staff. However, these signals may not always reflect real-time track conditions, especially if a manual check is still pending. In emergency situations, such delays can prevent timely decisions and actions by the pilot, compromising passenger safety.

### **3.Lack of Real-Time Automation and Alert Mechanisms**

Many current systems do not provide automated, real-time feedback about the state of the switch rails. Without intelligent sensors or data transmission to the locomotive or central control, the entire process remains reactive rather than proactive. There's no mechanism to alert the pilot or control centre immediately if the track alignment is incorrect or if the switch fails to lock into position.

## **Consequences of Track Alignment Failures:**

The consequences of even a single failure in track alignment detection can be catastrophic:

- **Derailments:**

A misaligned switch can lead a train to run off the tracks entirely. This is especially dangerous when the train is moving at high speed, as the force of the derailment can cause significant physical damage and lead to cascading failures in nearby infrastructure.

- **Delays in Train Schedules:**

Even if an accident is narrowly avoided, a malfunctioning or uncertain switch can bring entire sections of the network to a halt. Trains may be held back until manual verification is completed, leading to service disruptions, cascading delays, and financial losses for rail operators.

- **Accidents Resulting in Injury or Death:**

Tragically, history has shown that failures in switch systems can cause severe accidents, including collisions and derailments that result in loss of human lives, injuries to passengers and staff, and destruction of rolling stock.

- **Damage to Railway Infrastructure:**

A train that derails due to a faulty switch not only damages itself but can also destroy portions of the track, signalling systems, and nearby equipment. The cost of repairs and restoration can run into millions and take weeks or months, further disrupting operations.

### **Need for a Solution:**

Given these significant risks, there is an urgent need to modernize point switch systems with intelligent, real-time detection and communication capabilities. By implementing automated sensors, wireless communication modules (e.g., LoRa), and clear visual or voice alerts for train pilots, railway authorities can bridge the current gap in safety and responsiveness.

A smart switch detection system that can identify the alignment status of switch rails, communicate this wirelessly to the approaching train, and warn the pilot in real-time can dramatically reduce the chances of derailment, improve safety, and enhance operational efficiency.

#### 4. Objectives

| Objective ID | Description                                                           |
|--------------|-----------------------------------------------------------------------|
| OBJ-01       | Detect the alignment status of the railway switch rails using sensors |
| OBJ-02       | Transmit the switch status wirelessly to a locomotive receiver        |
| OBJ-03       | Display clear messages to the train pilot regarding track status      |
| OBJ-04       | Establish reliable communication when the train is approx. 1 km away  |
| OBJ-05       | Reduce the chance of derailments using early alerting                 |

## 5. Literature Review

### Key Findings from Existing Literature and Reports

#### 1. Indian Railways Safety Report (2021)

One of the most eye-opening revelations came from the **Indian Railways Safety Report of 2021**, which provided comprehensive data on accidents and their causes across the Indian railway network. According to this report, **34% of all derailments reported were directly linked to point switch failures**. This figure is not only alarming but also reflects a systemic issue that continues to threaten the safety and efficiency of rail travel.

The report emphasized that most of these derailments occurred because of improperly set or misaligned switch rails. In many instances, the root cause was traced back to **human error in verifying the switch position or in operating manual lever systems**. These failures highlight the inherent risk of relying on manual oversight in a system where timing and precision are critical. The report further called for modernization and stricter maintenance schedules but stopped short of recommending real-time alerting or wireless notification systems for train operators.

#### 2. IEEE Journal on Railway Systems (2020)

This paper discussed the design and implementation of **electromechanical and sensor-based systems** to automate switch position detection. These technologies aimed to reduce dependence on manual intervention and improve accuracy in determining the status of the switch rails. While the proposed systems in the IEEE study showed promise, especially in terms of **automated fault detection and real-time feedback to control centers**, they did not fully address one of the most critical needs: **direct communication with the train pilot**. The study lacked an integration layer that could send real-time alerts or visual/audio messages to the pilot inside the locomotive, especially in remote areas where track visibility is limited. As a result, the pilot remains uninformed if a track alignment failure occurs until it's potentially too late. This gap illustrates that while automation is progressing, the human-machine interface, particularly at the driver level, is still underdeveloped.

#### 3. IRCTC Technical Whitepaper

The **IRCTC (Indian Railway Catering and Tourism Corporation) Technical Whitepaper** provides a practical, ground-level perspective on how point switches are currently managed in

many parts of India. It outlines the standard operating procedures, including **routine manual inspections and physical verification by railway staff**. The whitepaper acknowledged that while these practices have worked to an extent, they are no longer sufficient in today's high-speed and high-density rail environment.

One of the key issues raised was the **inconsistency of manual inspections**. Track maintenance staff often work under extreme weather conditions, over long shifts, and are sometimes required to verify multiple switch points in a short period. This not only increases the likelihood of human error but also leads to fatigue-related oversights. Additionally, the whitepaper noted that in rural and semi-urban regions, the communication between trackside teams and the control center is often delayed due to lack of connectivity or outdated equipment.

The IRCTC document strongly advocated for the introduction of **remote monitoring and predictive maintenance tools**, including sensor-based systems. However, similar to the IEEE study, it did not propose solutions for **alerting the train pilot directly** or integrating wireless modules for on-board notifications, which leaves a critical gap in the safety framework.

### **Conclusion from Literature Review**

The insights drawn from these key sources make it abundantly clear that while awareness of the problem exists, and automation efforts have begun, **current systems still lack holistic, real-time solutions that involve the train pilot in the safety loop**. A truly effective system must not only detect switch alignment but must also **immediately communicate the status wirelessly to the approaching locomotive**, providing visual or audio alerts to the pilot before any hazard occurs.

These findings strengthen the motivation behind developing a **smart, sensor-based point switch detection and alert system**—a system that bridges the gap between the track infrastructure and the train cabin to prevent derailments, delays, and potentially fatal accidents.

**Gap Identified:** Most systems do not integrate **wireless pilot alerts** or **automated real-time detection** based on loop track alignment.

## **6. Methodology**

### **6.1 Overview**

To address the critical safety challenges in existing rail switching systems, our proposed solution is designed to automate the detection of track alignment and wirelessly alert the locomotive pilot in real time. This methodology leverages modern sensor technology and long-range wireless communication to ensure that the train pilot is informed well in advance if the track is improperly aligned.

The system consists of four key components working in an integrated manner:

#### **1. Inductive Proximity Sensors for Track Alignment Detection**

At the heart of the system are inductive proximity sensors, which are known for their reliability, durability, and resistance to harsh outdoor conditions. These sensors are strategically installed near the point switch to detect the exact position of the movable switch rails.

- When the rails are correctly aligned and locked into position (either for the main track or the diverging track), the proximity sensor detects the presence of metal at the expected location and sends a digital signal indicating proper alignment.
- If the rails are not properly set or stuck midway, the sensor fails to detect the expected signal, indicating misalignment.

This real-time detection capability removes the need for manual visual inspection and ensures that the track's condition is continuously monitored with millisecond-level response time.

#### **2. Transmitter Circuit for Processing Sensor Data**

The signals from the proximity sensors are fed into a microcontroller-based transmitter unit placed trackside near the switch assembly. This unit processes the sensor data and translates it into a meaningful status — either "Track Aligned" or "Track Not Aligned".

The transmitter unit is equipped with logic that:

- Continuously reads sensor outputs
- Determines whether the detected state is safe for train passage
- Prepares the corresponding status message to be sent wirelessly

Additionally, the transmitter unit includes indicators (e.g., LEDs or buzzers) for on-site maintenance teams to observe real-time status visually, if required.

### **3. Receiver Unit in the Locomotive Pilot Cabin**

To ensure that the train pilot is always aware of track alignment status before reaching a switch point, a receiver module is installed in the locomotive pilot's cabin. This unit is responsible for receiving the wireless signal transmitted from the trackside transmitter.

Upon receiving the data, the receiver:

- Displays the alignment status using a simple LED and screen-based interface (e.g., "Track Aligned" or "Track Not Aligned")
- Triggers a buzzer if the track is misaligned, thereby immediately notifying the pilot to slow down or stop
- Provides continuous feedback until the switch is cleared or aligned properly

This on-board notification system ensures that the pilot is not solely dependent on external signalling systems, which might be delayed or unclear.

### **4. LoRa Communication for Long-Range Wireless Transmission**

To connect the trackside transmitter and the on-board receiver, the system uses LoRa (Long Range) communication technology, which is ideal for transmitting low-bandwidth sensor data over distances up to 1 kilometre or more, even in challenging environments.

LoRa operates in unlicensed radio spectrum and is highly energy-efficient, making it suitable for remote railway environments where power sources may be limited. Its strong signal penetration also ensures that communication is maintained even with physical obstructions such as trees, poles, or small buildings.

The use of LoRa eliminates the need for wired communication infrastructure, reducing installation costs and simplifying deployment. The system can reliably transmit alignment data from the track to the moving train well in advance of the switch, giving the pilot sufficient time to react if necessary.

## Conclusion of Methodology Overview

This combination of real-time sensing, wireless data transmission, and in-cabin alerts creates a closed-loop safety system that significantly reduces the risks associated with manual track alignment checks. By automating detection and delivering alerts directly to the train pilot, our system aims to prevent derailments, enhance operational efficiency, and modernize traditional railway safety practices.

## 6.2 Step-by-Step Operation

### 1. Track Alignment Monitoring

The **track alignment monitoring** process begins with the installation of an **inductive proximity sensor** at a critical location near the **movable point rail**—specifically at the end of the **loop track** where it connects with the **main line**. This sensor plays a crucial role in continuously checking whether the movable rail (switch rail) has been correctly positioned to allow safe passage of the train. Proximity sensors are non-contact sensors capable of detecting metallic objects within a specified sensing range. In this context, the sensor is precisely aligned so that it detects the **metallic rail only when the track is correctly aligned**. If the rail is not in the expected position (for example, stuck halfway between two tracks or not locked properly), the sensor fails to detect the presence of metal, thus signalling a misalignment.

This mechanism ensures:

- **Real-time verification** of whether the switch rail is in the correct and safe position.
- **Continuous monitoring**, as the sensor can operate 24/7 regardless of weather conditions.
- **Maintenance-free operation**, since inductive sensors are durable and resistant to dirt, moisture, and vibration—making them ideal for rugged railway environments.

By placing the sensor strategically near the junction between the loop and main track, the system can reliably determine if the train can safely proceed or must be halted to avoid derailment.

## 2. Signal Transmission

If aligned, the sensor triggers the **transmitter circuit**. Transmitter sends a **digital signal via LoRa** module.

## 3. Receiver Functionality

When the locomotive is approximately **1 km** away:

The **receiver establishes a connection** and displays "**Waiting for Signal**".

If signal is received:

It displays "**Track Detected**" and turns on **green LED**.

If signal is not received within a certain time:

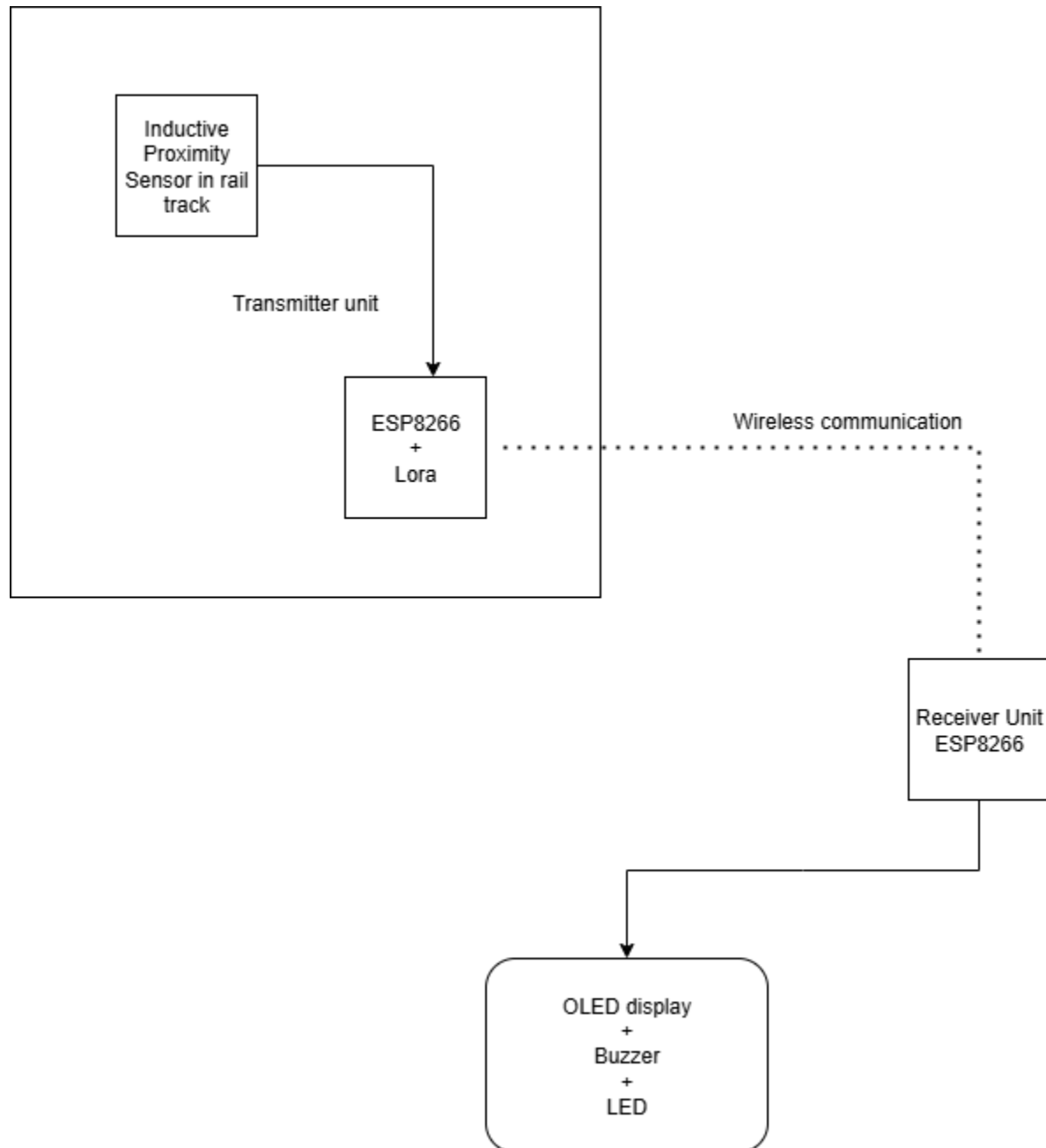
It displays "**Track Not Detected**" and turns on **red LED**.

## 7. Components Used

| Component                  | Specification            | Purpose                                       |
|----------------------------|--------------------------|-----------------------------------------------|
| Inductive Proximity Sensor | 5V, NPN type             | Detects alignment of track rails              |
| Servo Motor                | 12V                      | (For demonstration only) switch movement      |
| LoRa Module                | SX1278 433 MHz           | Long-range wireless communication             |
| Microcontroller            | ESP8266                  | Logic control for both transmitter & receiver |
| OLED Display               | 0.96" I2C                | Message display in locomotive cabin           |
| LEDs                       | Red and Green            | Visual indication for pilot                   |
| Power Supply               | 12V DC                   | Supply for sensors, microcontroller, etc.     |
| Misc.                      | Wires, Board, Connectors | Circuit completion                            |

## 8. System Architecture

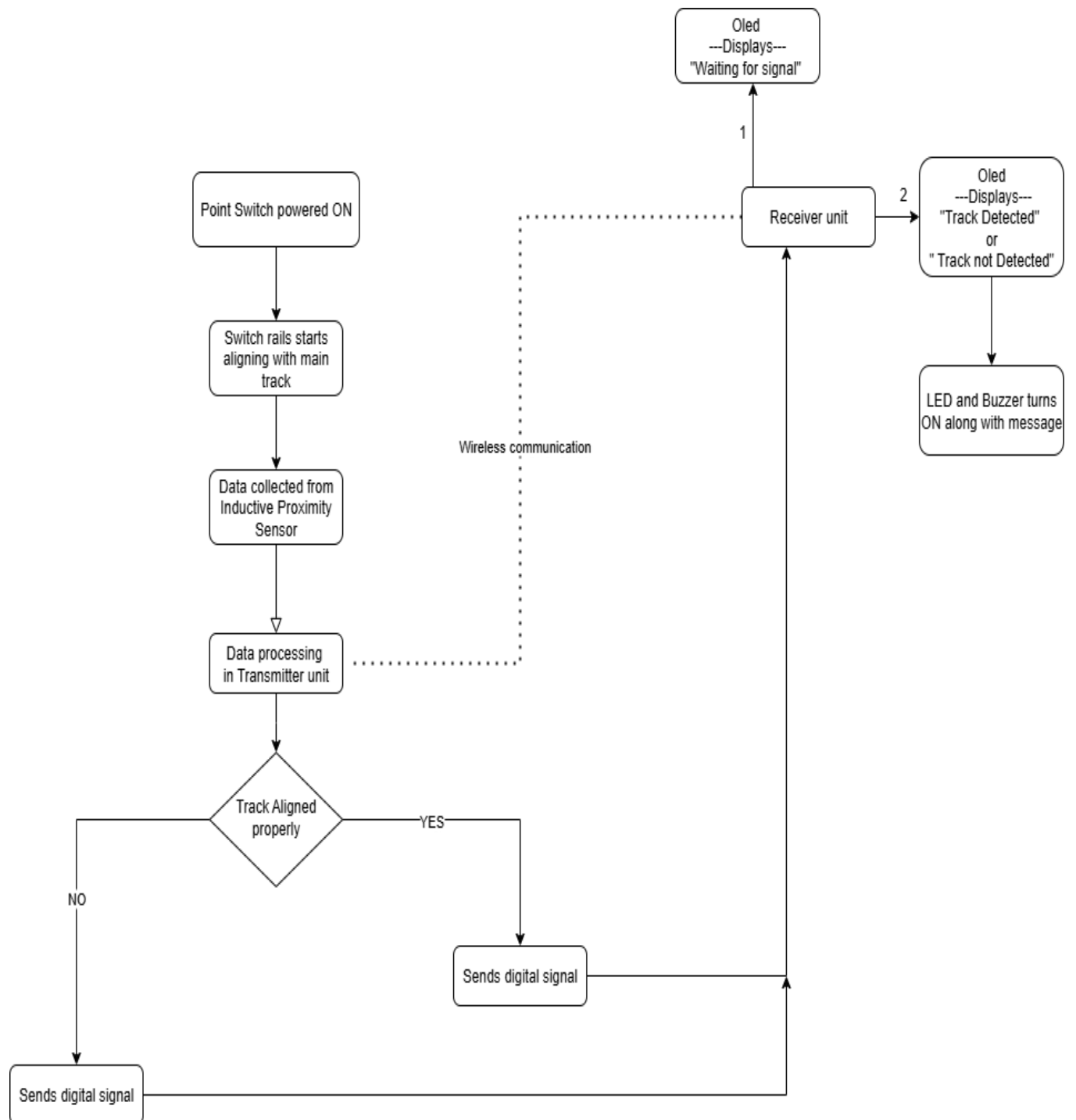
### 8.1 Block Diagram



## 9. Working Principle

1. When the switch moves, the **loop track aligns** with the mainline.
2. The **proximity sensor** detects this alignment.
3. If aligned, the sensor outputs a signal to the **transmitter unit**.
4. When the train is **within 1 km**, the **receiver activates** and displays **“Waiting for Signal.”**
5. Upon receiving signal:  
**Display shows “Track Detected”**
6. If signal is not received in time:  
**Display shows “Track Not Detected”**

## Flowchart:



## 10. Results & Observations

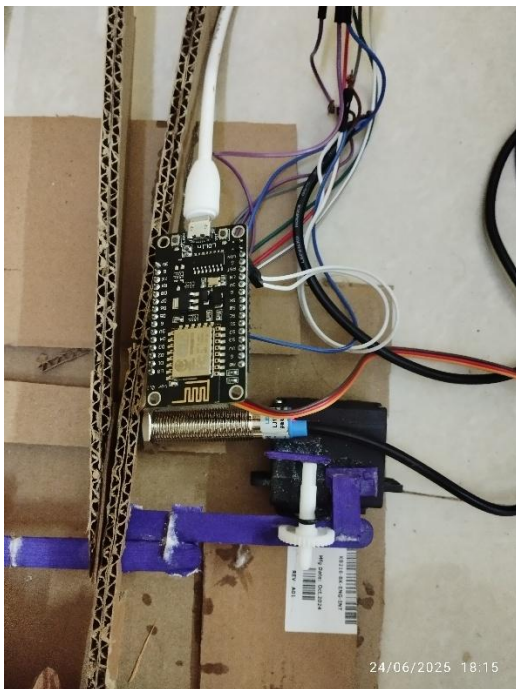
| Test Scenario                              | Expected Outcome               | Actual Outcome                               |
|--------------------------------------------|--------------------------------|----------------------------------------------|
| Proper track alignment                     | “Track Detected” + LED         | <input checked="" type="checkbox"/> Achieved |
| Improper or no alignment                   | “Track Not Detected” + LED     | <input checked="" type="checkbox"/> Achieved |
| Post- wireless communication establishment | “Waiting for Signal” initially | <input checked="" type="checkbox"/> Achieved |
| Receiver out of range                      | No update, default safe state  | <input checked="" type="checkbox"/> Achieved |
| Sensor disconnected                        | "Track Not Detected" fallback  | <input checked="" type="checkbox"/> Achieved |



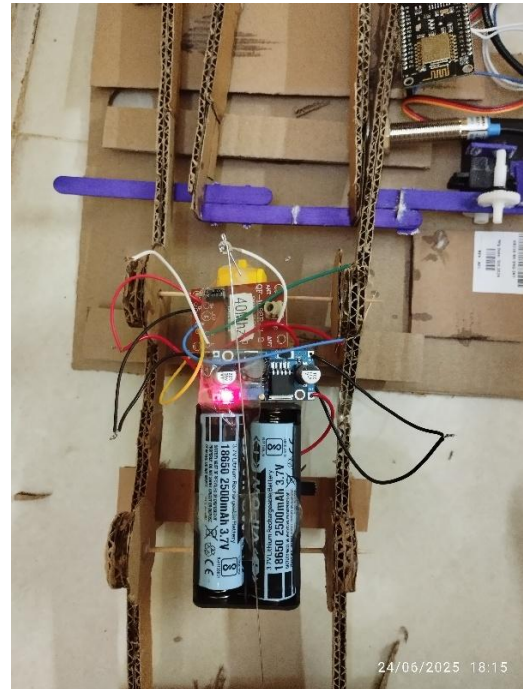
**Fig:10.3 OLED shows  
“Track Detected”**



**Fig:10.4 OLED shows  
“Track not Detected”**



**Fig:10.5 Inductive Proximity  
Sensor with ESP8266**



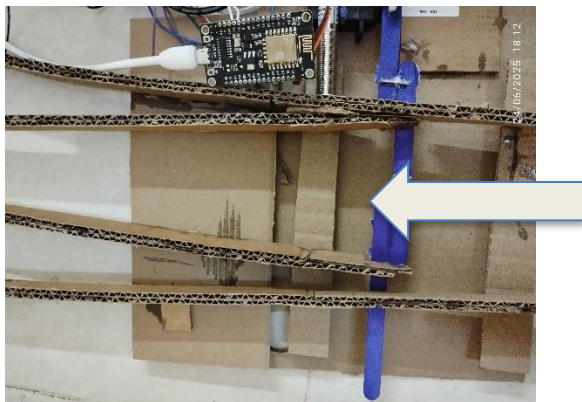
**Fig:10.6 Testing Our sample  
Loco motive**



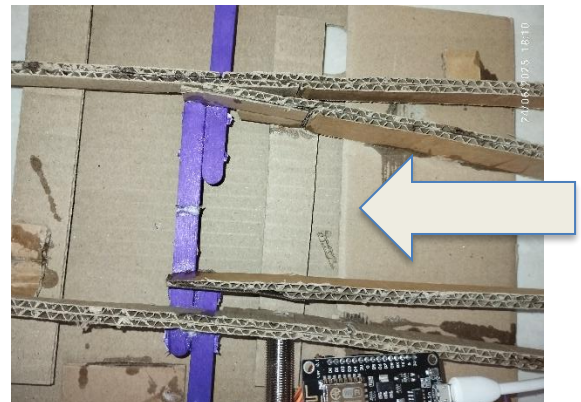
**Fig:10.7 Model of Track  
Diversion**



**Fig:10.8 OLED shows “Waiting for  
Signal” message after transmitter  
connected with receiver**



**Fig:10.1 Arrow indicates Facing  
Direction**



**Fig:10.2  
Arrow indicates Trailing Direction**

## 11. Advantages

The proposed point switch detection and alert system brings multiple advantages that directly address the shortcomings of traditional railway switching mechanisms. By integrating modern sensor technology with efficient wireless communication, the system enhances both **safety** and **operational efficiency** in a cost-effective manner. Below are the key benefits of this solution:

### 1. Cost-Effective and Scalable System

One of the most attractive features of the system is its **affordability and scalability**. The hardware components—inductive proximity sensors, microcontrollers, and LoRa modules—are all low-cost and readily available in the market. This makes the solution **financially viable for deployment even in resource-constrained railway networks**. Furthermore, the system’s modular design ensures that it can be easily installed at individual switch points without requiring significant changes to existing infrastructure. This makes it **scalable across large networks**, allowing operators to start small (such as at high-risk junctions) and expand over time.

### 2. Real-Time Notification for Train Pilots

A major limitation of conventional systems is the delay in communicating the status of track switches to the locomotive pilot. This system eliminates that gap by providing **instantaneous, real-time updates directly to the train pilot’s cabin**. If a track is misaligned or not detected properly, the system immediately triggers a **visual indicator (e.g., red light or screen message)** and optionally an **audible alert**, allowing the pilot to take corrective action well in advance. This direct feedback loop between the track and the train drastically reduces the risk of human oversight and delayed responses.

### 3. Reduces Accidents Caused by Manual Switch Errors

Many railway accidents, especially derailments, have been traced back to **human error in manual switch operations**—either due to failure in setting the switch correctly or in miscommunicating its status. By automating the detection of switch rail alignment and eliminating the dependency on manual verification, the proposed system **significantly reduces the risk of such accidents**. This not only improves safety but also enhances the confidence of passengers and crew in the system's reliability.

### 4. LoRa Provides Long-Range Communication with Low Power Consumption

The use of **LoRa (Long Range) technology** is a major advantage in this system. LoRa supports wireless data transmission over distances up to 1 kilometre or more in open conditions—**perfectly suited for railway environments**, especially in rural or remote areas. Despite its impressive range, LoRa operates with **extremely low power consumption**, making it ideal for battery-powered or solar-assisted setups. Additionally, LoRa is resilient to environmental interference and can operate in unlicensed spectrum bands, making it a **cost-efficient and technically robust** choice for railway communication.

### 5. Failsafe Design – Defaults to “Not Detected” if Communication Fails

Safety is at the core of the system's architecture. It has been designed with a **failsafe approach**, meaning that in case of any component failure—such as sensor malfunction, transmitter fault, or communication drop—the system automatically **defaults to a "Track Not Detected" status**. This conservative response ensures that the train pilot is alerted to a potential issue, prompting caution or a stop, rather than allowing the train to proceed based on unverified or missing data. This design philosophy greatly reduces the chances of undetected system failure leading to an accident.

## **12. Limitations & Challenges**

While the proposed rail switch detection and alert system introduces significant improvements in safety and automation, it is important to acknowledge that, like any technological solution, it comes with certain limitations and implementation challenges. Understanding these challenges helps in designing better mitigation strategies and planning for future enhancements. The following are the key concerns identified during the system's conceptualization and prototyping stages:

### **1. Environmental Interference**

One of the main challenges in deploying inductive proximity sensors on railway tracks is exposure to harsh environmental conditions. The track area is constantly subjected to dust, oil leaks from locomotives, rain, humidity, mud splashes, and in some regions, even snow or extreme heat. While inductive sensors are generally rugged and weather-resistant, excessive accumulation of dirt or debris can occasionally impact their ability to reliably detect the presence of metal (i.e., the rail). In some cases, if the sensor's face is covered or blocked, it may give false-negative readings, mistakenly identifying the track as misaligned. This could lead to unnecessary alerts or halts. Therefore, regular maintenance and possible inclusion of sensor shields or self-cleaning mechanisms may be required to maintain long-term reliability in all-weather conditions.

### **2. Power Requirements in Remote Areas**

The system's trackside components—especially the proximity sensor, transmitter circuit, and LoRa module—require a consistent power supply to function effectively. In urban and semi-urban areas, this may not pose a problem due to access to the railway's electrical grid. However, in remote or rural areas, continuous power availability can become a major hurdle.

In such scenarios, the system may need to rely on solar-powered energy sources combined with battery storage to ensure uninterrupted operation, especially during cloudy or rainy days. This adds some complexity to the design and may slightly increase the initial setup cost. Additionally, solar systems require occasional cleaning and protection from vandalism or wildlife interference, which may introduce minor operational overheads.

### **3. LoRa Signal Interruption**

Although LoRa is well-known for its long-range, low-power, and robust performance, it can still face signal attenuation and interference, especially in environments where there are large physical obstacles. This includes:

- Tunnels
- Underground tracks
- Densely built-up areas (stations with metal structures)
- Heavy rainfall or electrical interference zones

In such areas, the wireless communication between the trackside transmitter and the pilot's in-cabin receiver might experience delays or dropouts, reducing the system's real-time effectiveness. Careful placement of antennas, possibly using LoRa repeaters, and incorporating signal strength feedback mechanisms into the system can help address this limitation. Nonetheless, this challenge requires additional engineering consideration during deployment planning.

### **4. Installation Cost and Scalability**

While the individual components of the system—such as proximity sensors, microcontrollers, and LoRa modules—are relatively low in cost, scaling the solution across an entire rail network (especially a large one like Indian Railways) requires significant planning. Installation at thousands of points, coordination with track maintenance teams, safety approvals, and long-term maintenance agreements can lead to moderate capital and logistical investments.

Furthermore, the need to integrate this system with existing railway infrastructure and operations may call for customization at different locations, depending on rail geometry, climate, and available power. As such, even though the system is cost-effective at a small scale, a phased or prioritized deployment approach (starting with high-risk or high-traffic zones) may be necessary to manage budgets and resources efficiently.

### **13. Future Scope**

The proposed system holds immense potential for future enhancements that can significantly improve its intelligence, coverage, and interactivity. One promising upgrade is the integration of a high-definition camera module at the track switch, which, when combined with AI and machine learning algorithms, can visually validate the switch position and automatically detect obstructions or misalignments. Another valuable addition would be a mobile or web-based dashboard application that allows control room operators and train pilots to monitor real-time switch statuses across multiple junctions, improving coordination and situational awareness. GPS integration can also play a critical role by mapping the train's live location in relation to upcoming switch points, enabling context-aware alerts and improving navigation. Enhancing the system with two-way communication would allow train pilots to acknowledge messages or report faults directly back to the control center, closing the feedback loop and ensuring faster response times. Lastly, implementing a data logging feature to track switch status history, alignment events, and sensor behavior over time would enable predictive maintenance, allowing railway authorities to proactively address potential failures before they impact operations. These future developments would elevate the system from a basic detection tool to a fully intelligent, networked safety platform for modern railways.

## 14. Conclusion

This project introduces a **reliable, wireless, sensor-based solution** for detecting railway point switch functionality. By leveraging **Inductive proximity sensors** and **LoRa-based communication**, it eliminates manual dependency and gives real-time status updates to the loco-pilot. The system significantly enhances safety, is affordable for mass deployment, and is adaptable for future integrations like vision-based analysis and remote dashboards.

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